

CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

Project Report

Submitted By:

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Certificate

This is to certify that the project titled “**Customer Segmentation & Churn Pattern Analytics in European Banking**” is a bonafide work carried out by **Raman Pratap Singh** in partial fulfillment of the academic requirements.

Declaration

I hereby declare that this project is my original work and has not been submitted elsewhere for any degree or diploma.

Signature: Raman Singh

Acknowledgement

I would like to express my sincere gratitude to my faculty guide and institution for their continuous support and guidance throughout this project.

1. Executive Summary

Customer churn is a major challenge in retail banking. This project analyzes customer behavior, identifies meaningful customer segments, and evaluates churn patterns using data analytics techniques. The study considers geographic, demographic, financial, engagement, credit, tenure, and product-related characteristics. An interactive dashboard is proposed to help decision-makers identify high-risk segments and support targeted retention strategies.

2. Introduction

In a competitive banking environment, retaining existing customers is important for long-term customer value and revenue stability. Customer churn occurs when customers discontinue their relationship with a bank. Instead of relying only on an overall churn rate, segmentation helps identify which groups are more vulnerable and which groups may represent greater financial exposure.

3. Problem Statement

Banks may possess extensive customer-level data but still face difficulty converting it into actionable retention strategies. A general churn rate does not reveal which countries, age groups, financial profiles, or engagement segments are most at risk. This project addresses this gap through systematic customer segmentation and churn analytics.

4. Objectives

Primary objectives:

- Measure overall churn rate.
- Identify customer segments with higher churn rates.
- Compare churn across European countries.
- Analyze demographic and financial churn patterns.
- Identify high-value customers who have churned.
- Develop an interactive analytics dashboard.

Secondary objectives:

- Understand engagement and tenure patterns.
- Examine balance and product relationships.
- Support targeted customer-retention decisions.

5. Dataset Description

The dataset contains customer-level banking information. Key variables include CustomerId, Surname, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited. The Exited variable is used as the churn indicator.

6. Data Cleaning & Preparation

The preparation stage includes checking duplicates and missing values, validating binary fields such as HasCrCard, IsActiveMember and Exited, excluding Surname from analytical calculations, and standardizing categorical variables for grouping and comparison.

7. Methodology

The project follows a structured analytics process: data ingestion and validation, cleaning and preparation, customer segmentation, exploratory data analysis, KPI calculation, visualization, business interpretation, and recommendation development.

8. Customer Segmentation Analysis

Customers are segmented by geography (France, Spain, Germany), age (<30, 30–45, 46–60, 60+), credit score bands, tenure groups, balance segments, and engagement status. These segments allow churn behavior to be compared across customer profiles.

9. Churn Analysis

Churn analysis includes overall churn rate, segment-wise churn rate, churn contribution by segment, and comparisons between churned and retained customers. Important dimensions include geography, age, gender, tenure, credit score, balance, number of products, and customer activity.

10. Dashboard Insights

The Streamlit dashboard should present overall churn KPIs, geography-wise churn, age and tenure comparisons, product analysis, activity-based churn, balance analysis, and a high-value customer explorer. Interactive filters should dynamically update the KPIs and visualizations.

11. Key Performance Indicators (KPIs)

Key KPIs include Total Customers, Churned Customers, Overall Churn Rate, Segment Churn Rate, High-Value Customer Churn Ratio, Average Balance, Average Salary, Average Tenure, and Engagement Drop Indicator.

Overall Churn Rate = Churned Customers / Total Customers × 100

12. Business Insights

The final business insights should be generated from the actual dashboard results. Areas of focus include geographic risk, engagement-related churn, high-value customer churn, product relationships, age-based risk, and tenure patterns. Numerical findings should be inserted after the dashboard analysis is completed rather than assumed in advance.

13. Recommendations

Banks can consider targeted retention campaigns for high-risk segments, re-engagement strategies for inactive customers, priority treatment for high-value customers, country-specific retention strategies, suitable product-based engagement, and an early-warning framework for customers showing characteristics associated with higher churn risk.

14. Limitations

The dataset may not contain customer satisfaction, transaction behavior, competitor information, or explicit reasons for churn. The analysis identifies associations rather than proving causation. Definitions of high-value customers may also differ by banking strategy.

15. Future Scope

The project can be extended through predictive churn modeling using Logistic Regression, Decision Trees, Random Forest or Gradient Boosting. Future work may also integrate Customer Lifetime Value, K-Means clustering, real-time analytics, and AI-based personalization.

16. Conclusion

Customer churn analytics can help banks move beyond simply measuring how many customers leave toward understanding who leaves, which segments are vulnerable, how financially important those customers are, and what targeted actions can improve retention. The project demonstrates how customer-level banking data can be transformed into business intelligence through segmentation, KPI analysis, visualization, and interactive dashboards.

17. References

- European Central Bank (ECB) – Banking and financial sector information.
- Unified Mentor – Project and learning resources.
- Streamlit Documentation – Dashboard and application development resources.
- Pandas Documentation – Data manipulation and analysis resources.
- Matplotlib/Plotly Documentation – Data visualization resources.
- Kaggle banking customer churn dataset resources.